

Final Project Rubric

In total projects are graded out of 80 points: 8 points for the proposal, 42 points for the presentation, and 30 points for the report.

Presentation

Your presentation will be 10-12 minutes with 5 minutes for questions. Presentations will be in-class on December 3, 8, and 10. All students should attend and be prepared to ask questions, as it will contribute to their grade. Presentations are worth 42 points total. **Students may not use notes during their presentations.** For projects with two students, both students must answer substantive questions and will be graded separately on answering questions.

You will be evaluated on each of the following components. You are not bound to the following order, but you may find it to be a convenient order.

1. Motivation (3 points)
2. High-level research question (3 points)
3. Precise model (3 points)
4. Statement of main results (4 points)
5. Contextualization of results
 - (a) with respect to related work (3 points)
 - (b) the significance of this contribution (3 points)
6. Additional preliminary mathematical building blocks necessary to understand (3 points)
7. Proof sketches of main results **focused on intuition** and any key/cute/novel ideas. (i.e. not just calculations) (3 points)
8. Open problems that you like (3 points)

Advice: The model and statements should be clear, but also try not to use too much math/notation. It is a good idea to use running examples!

Finally, students will be evaluated on:

- Quality of presentation: running examples, clear model and statements, not too much notation, general clarity. (3 points)
- Length of presentation (3 points)

- Demonstrated understanding in answering questions: explaining assumptions, proof ideas, limitations, and your contribution. (5 points)
- Participation in other presentations (3 points)

Write-Up

Your write-up should be typeset in LaTeX and be **at least 6 pages long, not including references**. It must be turned into Gradescope by 5pm on December 19, one copy per group. **No late reports will be accepted.** It is worth 30 points total.

Research Write-Up: You will be evaluated on each of the following components.

1. Introduction: high-level research question (6 points), motivation (2 points), statements of main results (2 points), a subsection on related work (2 points)
2. Preliminaries: precise model, additional preliminary mathematical building blocks necessary to understand the remainder of the work (2 points)
3. Bulk of the write up split into sections that tell a story, including:
 - Results (4 points)
 - Formal theorem statements (2 points), formal proofs (2 points), english paragraphs of intuition and telling a story in between them (2 points)
 - Your approach and results, discussion of challenges faces, other approaches attempted (2 points)
 - Suggestions for future work (2 points)
4. References—always be generous with your citations. (2 points)

Research write up example: <https://arxiv.org/pdf/2011.09406.pdf>